Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/17/2025
PI Name:	Rosen, Michael Ph.D.		
Project Title:	Developing and Validating Sensor-based Measurement Strategies for Team Member Selection		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21202-3142	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
Start Date:	12/01/2016	**End Date:**	05/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2024 per A. Beitman/NASA JSC (Ed., 11/15/23) NOTE: End date changed to 11/30/2023 per A. Beitman/NASA JSC (Ed., 10/20/2022) NOTE: End date changed to 11/30/2022 per NSSC information (Ed., 11/2/2021) NOTE: End date changed to 11/30/2021 per NSSC information (Ed., 10/23/2020) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 1/30/2020)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI): Drs. Sapirstien and Wick have left the project. A pediatric ICU (PICU) will be used instead of a surgical intensive care unit and neither Drs. Sapirstien nor Wick have involvement there. PICU faculty did not need support. (Ed., 2/15/23). January, 2023: Molly Kilcullen and Jaxon Wu have been added to the project to assist with data analysis in the final period. October 2021 report: Dr. Dietz and Nam Lee have left Johns Hopkins University (JHU) and are no longer working on the project.		

COI Name (Institution):	Oswald, Fred Ph.D. (Rice University) Salas, Eduardo Ph.D. (Rice University)
Grant/Contract No.:	NNX17AB55G
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Task Description:	Selection of astronauts for Long Duration Spaceflight Exploration (LSDE) missions poses challenges for NASA including the need to define and select candidates based on a new set of behavioral competencies underpinning effective performance in these extended and isolated missions. Additionally, an effective selection system will require new measurement methods capable of discriminating between individuals in a population already exhibiting extreme range restriction. Sensor-based, sociometric, and more generally, unobtrusive measurement methods hold promise as valuable tools for addressing these needs and complementing existing competency assessment methods. The proposed work seeks to advance the science and practice surrounding diagnostic measurement of LDSE competencies using a blended approach where sociometric techniques are combined with traditional assessment methods. We will leverage our team's extensive, transdisciplinary experience in signal processing and analysis of complex dynamic network data, psychometrics, performance assessment, and developing theory and strategies for LDSE team improvement to: (1) generate predictive validity evidence for LDSE behavioral competencies, (2) develop sociometric indices of those competencies and provide evidence of their validity, (3) develop an open architecture system for integrating sensor-based measurement systems and extracting sociometric indices, and (4) generating guidelines for the use of sociometric measures in the selection process. Our technical approach for achieving these aims involves metric development, metric validation, assessment architecture system design, and selection guideline development. First, metric development will involve updating our current literature review of unobtrusive measurement to incorporate findings from recent NASA efforts. We will also apply reactive systems modeling to systematically map sensor-based measurement system requirements with potential metrics for assessment. Next, we will conduct exploratory human in the loop analyses to identify additional candidate measures using tensor-decomposition methods of archival data to detect performance patterns. Metric validation will occur in a LDSE analog (HERA--Human Exploration Research Analog) as well as two clinical residency programs in order to increase the sample size needed for analysis. Specifically, we will prospectively collect the following from each context: traditional assessments of competencies (self-report, observation), sociometric assessments of competencies, and multi-dimensional outcomes (task outcome measures, multiple rater assessments). Validation studies will be conducted to establish the link between LDSE competencies and performance outcomes, demonstrate the relationships between sociometric and traditional measures of LDSE competencies, as well as index the amount of variability in outcome measures accounted for by sociometric indices above and beyond traditional measures. The next phase of this project involves advancing methodology and analytic capabilities for competency assessment using sociometric indices. The analytics developed will distill meaningful metrics from complex, dynamic sensor data that can be used across a range of sensor devices, providing a generalizable ‘middle layer’ architecture for processing these data. This analytic process involves integrating and mapping these diverse measurement sources to generate a valid and actionable depiction of performance (i.e., to guide selection). Findings from these efforts will result in evidence-based, practical, and validated guidelines for incorporating unobtrusive measurement into the astronaut selection process. Overall, successful completion of this project will advance the science and practice of multi-method individual and team LDSE competency assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Final products from this work will advance evidence-based selection practices for currently difficult to assess teamwork-related competencies. This will result in an astronaut corps more prepared to meet the demands of long duration space exploration. Additionally, these practices can benefit other professions where these competencies are germane.
Task Progress:	Literature reviews conducted early in the project identified four candidate unobtrusive measurement approaches: Linguistic (i.e., approaches for automated analyses of team communication), Paralinguistic (i.e., interpersonal communication not through language use, such as shared gaze, vocal tone, body movements), team physiological dynamics (i.e., TPD; mutual influence of team members’ physiology), and activity tracking (i.e., use of information systems to capture information about location, and task performance). Overall, the TPD profiling approach and crew member classes solve several challenges through unobtrusive measures for selection. High likelihood of utility in selection systems: The profiling approach shows validity for identifying differences in individuals that rated to team outcomes, and traits known to be important for team performance. Future development should address: 1) larger samples, in both isolated and confined environment (ICE) analogs, and individuals (re-) assembled in multiple teams; 2) tasks similar in structure to selection system. Linguistic style measures are (most likely) situational indicators of interpersonal dynamics. Not likely valid measures for selection: Conversational, turn by turn measures of matching and accommodation predict perceptions of speech similarity, but in complex ways.
Bibliography Type:	Description: (Last Updated: 11/25/2023)